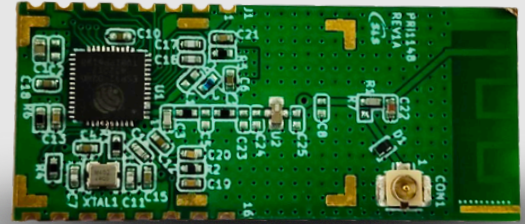


NM0442-P (Wi-Fi Module)

The NM0442-P Wi-Fi Module is a compact, high-performance wireless communication solution designed for embedded applications. Operating in the 2.4 GHz frequency band, it supports IEEE 802.11 b/g/n standards with data rates up to 150 Mbps. The module integrates a powerful Xtensa® LX6 dual-core 32-bit processor, large on-chip memory resources, and versatile interfaces, making it ideal for IoT and connected device applications. With features such as OTA (Over-the-Air) firmware updates, u.FL connector/PCB MiFA antenna options, and extended operating temperature support (-40°C to +105°C), the NM0442-P offers flexibility, scalability, and reliability for next-generation wireless devices.



Specifications

Performance			
TX Power		17dBm	
RX Sensitivity	Data Rate	1 Mbps. Sensitivity	-95dBm
		54 Mbps. Sensitivity	-75dBm
RF Data Rate		1 Mbps to 54 Mbps	
Processor Specifications			
Processor		Xtensa® single-/dual-core 32-bit LX6 microprocessor	
Memory		448 kB ROM, 520 kB SRAM, 16 KB SRAM	
Frequency Band		2.4GHz	
Antenna Option		u.FL Connector/PCB MiFA Antenna	
Size (in mm)		37.47x16 mm	
IO Specifications			
Serial Interface		1 x UART	
Analog Input		1 x ADC	
Digital IO		9 x Digital IO	
Electrical Specifications			
Supply Voltage		2.3V – 3.6V	
Current Consumptions		27~34mA	
Modem Sleep			
Light Sleep		0.8mA	
Transmit 802.11b		(DSSS 1 Mbps, POUT = +19.5 dBm)	240mA
Transmit 802.11g		(OFDM 54 Mbps, POUT = +16 dBm)	190mA
Transmit 802.11n		(OFDM MCS7, POUT = +14 dBm)	180mA
Operating Temperature		-40° C to 105° C	
Storage Temperature		-40° C to 150° C	

Applications



Industrial
IoT



Sensor &
gateways



Smart
Home



Consumer
Electronics

Key Product Features

- Wireless communication module for use in embedded applications
- Two timer groups, including 2 × 64-bit timers and 1 × main watchdog in each group
- 2.4GHz frequency band
- Memory Up to 448 kB ROM, Up to 520 kB SRAM
- 802.11 b/g/n, and 802.11 n (2.4 GHz), up to 150 Mbps compliant
- 32-bit Real Time Counter
- RTC watchdog
- Provides an OTA (Over the Air) Firmware update mechanism
- Comes with u.FL Connector/PCB MiFA Antenna selection option variant

Contact us

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